

CHARLO, NB, Canada

WMO: 713150

Lat: 47.9889N

Lon: 66.3336W

Elev: 42

StdP: 100.82

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.3	(c) -22.0	(d) -30.1	(e) 0.2	(f) -23.6	(g) -27.5	(h) 0.3	(i) -21.4	(j) 12.2	(k) -8.2	(l) 11.2	(m) -9.2	(n) 3.6	(o) 260	(p) 0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.7	(c) 28.5	(d) 19.8	(e) 26.4	(f) 18.8	(g) 24.6	(h) 18.0	(i) 21.3	(j) 26.4	(k) 20.2	(l) 24.7	(m) 19.1	(n) 22.9	(o) 4.0	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 19.5	(b) 14.3	(c) 24.0	(d) 18.5	(e) 13.4	(f) 22.5	(g) 17.6	(h) 12.6	(i) 21.3	(j) 62.2	(k) 26.4	(l) 58.2	(m) 24.9	(n) 54.7	(o) 23.0	(p) 25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.6	9.3	8.3	-28.1	31.8	2.0	1.5	-29.6	32.9	-30.8	33.8	-31.9	34.6	-33.4	35.7
			-28.4	23.2	2.0	0.9	-29.8	23.9	-31.0	24.4	-32.1	24.8	-33.5	25.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-11.1	-9.9	-4.5	2.2	9.2	14.9	18.7	17.8	13.1	6.8	0.3	-6.5
	DBStd	11.17	6.18	5.93	5.34	3.51	3.66	3.72	2.87	2.88	3.73	3.73	4.65	5.40
	HDD10.0	2889	655	558	449	234	59	3	0	0	12	113	293	512
	HDD18.3	5202	913	792	707	483	284	116	30	44	162	358	542	771
	CDD10.0	816	0	0	0	1	35	150	269	242	104	13	1	0
	CDD18.3	87	0	0	0	0	1	12	41	28	4	0	0	0
	CDH23.3	687	0	0	0	1	26	134	282	201	42	2	0	0
	CDH26.7	156	0	0	0	0	5	34	59	49	9	0	0	0
Wind	WSAvg	3.6	4.0	4.1	4.1	4.0	3.7	3.4	2.9	2.8	3.0	3.6	3.6	3.9
Precipitation	PrecAvg	1112	94	68	88	88	90	89	99	108	89	91	95	113
	PrecMax	1338	181	133	188	196	177	195	210	160	176	205	249	191
	PrecMin	819	12	9	37	24	15	35	37	62	17	22	39	61
	PrecStd	156	35	30	41	44	43	36	43	29	44	46	42	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	5.9	11.3	17.7	26.9	30.4	30.8	31.0	27.7	21.6	14.9	6.1
		MCWB	3.0	4.1	7.1	9.4	15.3	19.6	20.9	21.2	20.1	15.0	11.0	5.0
	2%	DB	1.8	2.5	5.9	12.8	22.1	27.0	28.3	27.8	24.0	17.0	10.3	3.8
		MCWB	0.8	1.3	2.3	6.7	13.7	18.1	20.2	20.2	18.1	12.9	8.3	2.6
	5%	DB	0.0	0.6	3.9	9.7	18.8	24.3	26.3	25.4	21.5	14.2	8.1	2.2
		MCWB	-0.7	-0.6	1.1	5.1	12.1	16.8	19.1	18.8	16.7	10.9	6.4	1.0
	10%	DB	-1.9	-1.0	2.2	7.4	16.0	21.7	24.6	23.5	19.4	12.2	6.3	0.9
		MCWB	-2.7	-2.1	0.2	3.6	10.4	15.4	18.2	17.8	15.4	9.8	4.7	0.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.3	4.1	7.1	10.7	17.6	21.6	22.8	22.5	21.1	16.4	12.9	5.2
		MCDB	4.0	5.7	9.6	15.4	23.5	27.9	28.0	28.4	25.8	19.9	14.9	5.8
	2%	WB	0.9	1.5	3.0	7.2	14.9	19.5	21.5	21.0	19.1	13.4	8.9	2.8
		MCDB	1.7	2.3	5.4	11.7	19.8	24.6	26.3	26.2	22.7	16.1	9.9	3.6
	5%	WB	-0.8	-0.4	1.6	5.5	12.9	18.0	20.3	19.8	17.4	11.6	6.6	1.3
		MCDB	-0.1	0.4	3.2	8.9	17.6	22.0	24.7	24.1	20.6	13.6	7.7	1.9
	10%	WB	-2.7	-2.1	0.4	4.1	11.2	16.6	19.3	18.8	15.9	10.0	4.9	0.1
		MCDB	-1.9	-1.1	1.9	6.6	14.9	20.4	23.1	22.1	18.7	11.7	6.2	0.8
Mean Daily Temperature Range	5% DB	MDBR	8.4	8.9	8.6	7.9	10.0	10.0	9.7	9.9	9.8	7.6	6.3	6.8
		MCDBR	9.2	8.6	9.9	12.6	15.7	14.8	12.8	13.1	12.7	11.2	8.7	6.5
		MCWBR	8.9	7.6	7.2	7.7	8.7	7.6	6.4	6.8	7.8	7.7	7.4	6.2
	5% WB	MCDBR	9.1	7.9	8.7	11.2	13.5	11.7	10.7	11.1	11.0	10.1	8.1	6.2
		MCWBR	9.0	7.3	6.8	7.5	8.4	6.9	6.0	6.3	7.5	7.6	7.4	6.2
Clear-Sky Solar Irradiance	taub	0.272	0.280	0.318	0.419	0.366	0.384	0.425	0.393	0.366	0.335	0.299	0.274	
	taud	2.328	2.287	2.265	2.091	2.378	2.348	2.268	2.357	2.413	2.492	2.508	2.404	
	Ebn at Noon	806	883	897	828	897	878	834	847	839	813	780	762	
	Edh at Noon	71	92	112	148	116	120	129	113	97	77	61	59	
All-Sky Solar Radiation	RadAvg	1.29	2.16	3.37	4.39	5.04	5.46	5.35	4.66	3.58	2.14	1.16	0.88	
	RadStd	0.15	0.25	0.24	0.39	0.43	0.40	0.49	0.29	0.28	0.19	0.15	0.09	

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) +2.06	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A
(47) Regional (26 neighbors)	DBAvg	(d) N/A	(e) N/A	(f) +1.34	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A

Nomenclature: See separate page